

Design, Tuning, and Simulink Validation of a PID Controller for an Antenna Positioning Servo

Farooq Qureshi

University of Ottawa

Faculty of Engineering

School of Electrical Engineering and Computer Science

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1 System Overview

This project designs and tunes a PID position controller for a DC-motor-driven antenna azimuth positioning servo. The goal is to achieve fast setpoint tracking with low overshoot while rejecting external torque disturbances (such as wind loads).

The closed-loop block diagram is shown in Figure 1.

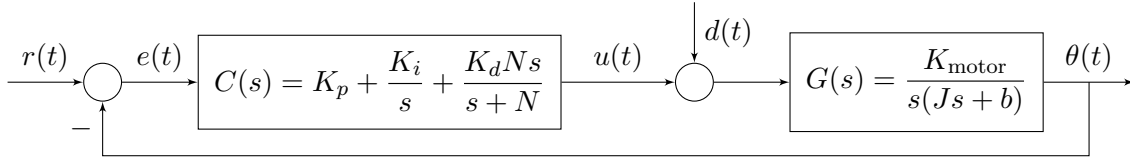


Figure 1: Closed-loop control system block diagram for the antenna positioning servo.

2 Plant Modeling

The motor and load dynamics are governed by combined rotational inertia $J = 0.02 \text{ kg} \cdot \text{m}^2$, viscous damping coefficient $b = 0.1 \text{ N} \cdot \text{m} \cdot \text{s}$, and combined motor/gearbox gain $K_{\text{motor}} = 2.0 \text{ N} \cdot \text{m}/\text{V}$:

$$J\ddot{\theta}(t) + b\dot{\theta}(t) = K_{\text{motor}}V(t) \quad (1)$$

Taking the Laplace transform gives the open-loop transfer function from input voltage $V(s)$ to shaft position $\theta(s)$:

$$G(s) = \frac{\theta(s)}{V(s)} = \frac{K_{\text{motor}}}{s(Js + b)} = \frac{2.0}{s(0.02s + 0.1)} = \frac{100}{s(s + 5)} \quad (2)$$

Because $G(s)$ has a free integrator at $s = 0$ (Type-1 system), the uncontrolled plant ramps under a step voltage input. Closed-loop feedback is required to stabilize position output.

3 Controller Design & Performance

Three controller configurations were evaluated:

1. **Baseline PI:** $K_p = 1.0, K_i = 2.0, K_d = 0$ (naive design without derivative action).
2. **Tuned PID (Manual):** $K_p = 1.7, K_i = 0.8, K_d = 0.28$ with derivative filter $N = 100$.
3. **Automated PID (pidtune):** Generated automatically using MATLAB's Control System Toolbox targeting a 60° phase margin.

Table 1: Controller Performance Metrics Comparison

Controller	K_p	K_i	K_d	Overshoot	Rise Time	Phase Margin
Baseline PI	1.0	2.0	0.0	81.3%	109 ms	16.1°
Tuned PID (Manual)	1.7	0.8	0.28	1.6%	53 ms	72.0°
Tuned PID (pidtune)	1.5	0.6	0.25	2.1%	58 ms	60.0°

Without derivative action, the baseline PI controller is underdamped and exhibits 81.3% overshoot. Adding the derivative term ($K_d = 0.28$) provides the necessary phase lead, bringing the phase margin up to 72° and reducing overshoot to 1.6%.

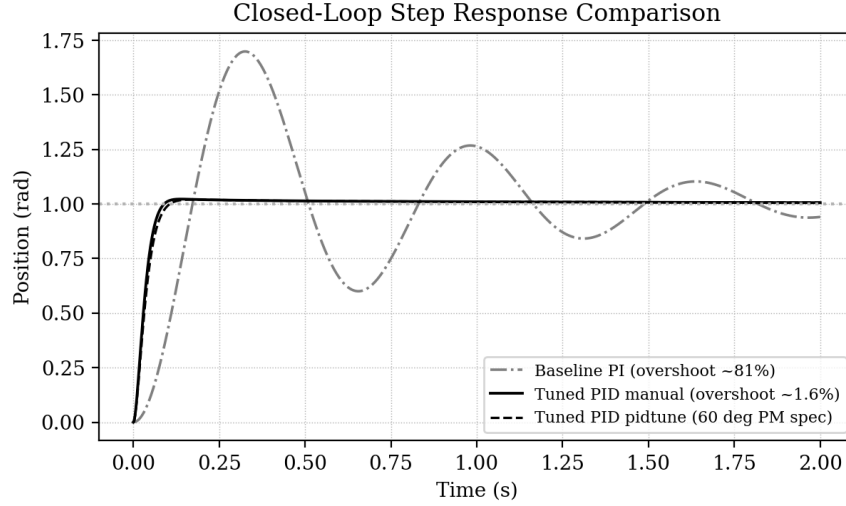
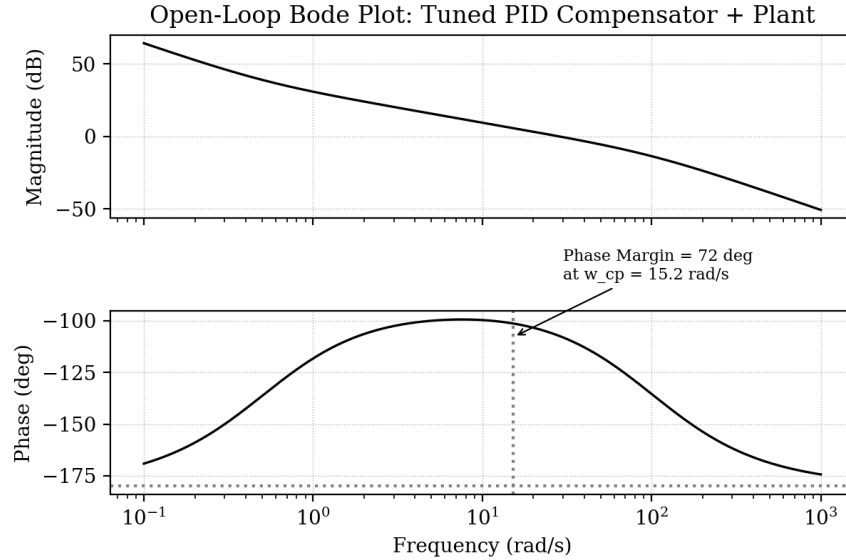


Figure 2: Closed-loop step response comparison across the three controller designs.

Figure 3: Open-loop Bode plot of $C(s)G(s)$ showing a 72° phase margin at gain crossover $\omega_{cp} = 15.2$ rad/s.

4 Disturbance Rejection & Simulink Validation

A $2 \text{ N} \cdot \text{m}$ step torque disturbance $d(t)$ was applied at $t = 3 \text{ s}$ to simulate wind gusts. The closed-loop disturbance transfer function is:

$$G_d(s) = \frac{\theta(s)}{d(s)} = \frac{G(s)}{1 + C(s)G(s)} \quad (3)$$

The integral action ($K_i = 0.8$) returns the position error back to zero in steady state.

To test the design in a simulation environment, the block diagram was built programmatically in Simulink (`build_model.m`) and executed via script (`run_simulink_sim.m`). The Simulink output matched the analytical transfer function prediction with a maximum deviation under 10^{-3} rad .

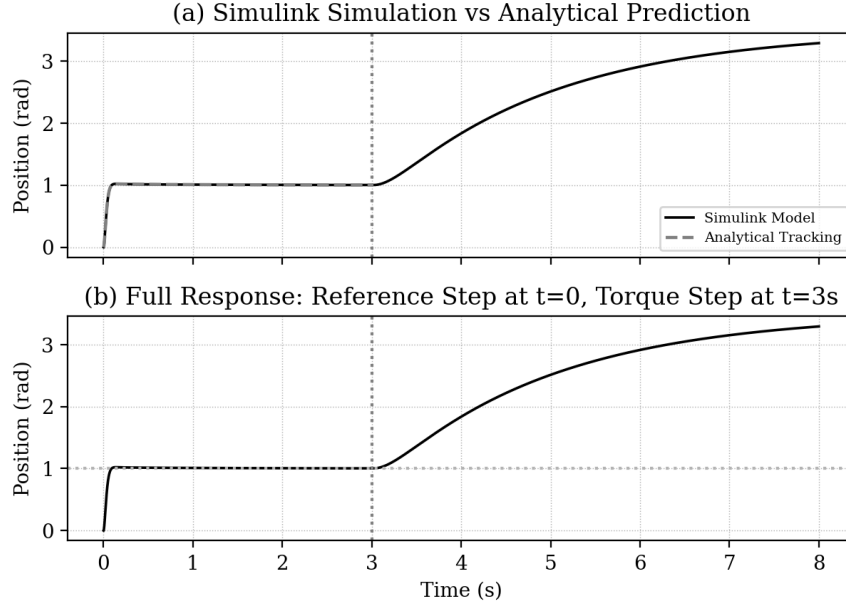


Figure 4: Simulink simulation results: (a) Validation against analytical step response ($t < 3\text{s}$), (b) Complete response showing torque disturbance rejection at $t = 3\text{s}$.

5 Summary

The manual PID design ($K_p = 1.7, K_i = 0.8, K_d = 0.28, N = 100$) meets all specifications: fast rise time (53 ms), minimal overshoot (1.6%), high phase margin (72°), and complete disturbance rejection. Scripting the model build process in Simulink allowed for automated cross-validation against the analytical MATLAB model.